

NILAY NAHA

+91 9749654480 ◇ Siliguri, West Bengal ◇ niloynaha2003@gmail.com ◇ LinkedIn — GitHub — Portfolio

PROFESSIONAL SUMMARY

Full Stack Developer — 2026 B.Tech CSE (AI & ML) graduate. Proficient in building scalable full-stack applications using React, Next.js, Node.js, Python, and cloud platforms, with hands-on experience in AI/ML pipelines and computer vision. Shipped multiple production-grade projects spanning SaaS platforms, real-time monitoring tools, and AI-powered solutions — several deployed and in active use. Built and maintained applications with measurable performance improvements including 40% faster page loads.

EDUCATION

B.Tech in CSE (AI & ML)

Siliguri Institute of Technology

2026 Graduate

CGPA: 7.58 (Final Year, Result Awaited)

Diploma in Computer Science & Technology

Siliguri Govt. Polytechnic

2020 – 2023

8.4 CGPA

TECHNICAL SKILLS

Frontend	React.js, Next.js, TypeScript, Tailwind CSS, HTML5, CSS3, JavaScript (ES6+), EJS
Backend	Node.js, Express, Python, Flask, REST APIs, Microservices, JWT Authentication
Database	MongoDB (NoSQL), SQLite (RDBMS), MySQL, Mongoose ODM
AI/ML	TensorFlow, OpenCV, XGBoost, SHAP, Scikit-learn, Pandas, NumPy, Neural Networks, Computer Vision
Cloud/DevOps	AWS, Docker, Vercel, Git/GitHub, CI/CD, Linux CLI
Core	OOP, DSA, SDLC, Agile, TDD, Debugging, System Architecture, API Design

CERTIFICATIONS

Google IT Automation with Python — Google

AWS Cloud Technical Essentials — AWS

Crash Course on Python — Google

Introduction to TensorFlow — DeepLearning.AI

Google AI Essentials — Google

Introduction to AI — Google

INDUSTRIAL EXPERIENCE

C-DAC (MeitY Project, Government of India)

Project Lead — 3D Printing & Additive Manufacturing

May 2025 – July 2025

Siliguri, WB

- Led a 4-member team, managed end-to-end project delivery, and achieved 100% operational uptime through systematic testing and troubleshooting.
- Authored a 39-page technical report documenting system architecture and operational procedures — reducing onboarding time for new team members by 30%.

KEY PROJECTS

ShortDesk — Smart URL Shortener

 GitHub

Node.js, Express, MongoDB, Redis

- Architected full-stack URL shortening platform with password-protected links, click analytics, and Google Safe Browsing API integration — deployed with 99.9% uptime on Vercel.
- Designed atomic Base62 ID generation (zero collisions) with Redis caching for non-blocking analytics and serverless deployment with MongoDB Atlas.

Full-Stack Portfolio & Dynamic Blog

 GitHub

Next.js 14, TypeScript, MongoDB, Tailwind CSS

- Built full-stack blog platform with Next.js App Router, TypeScript, and MongoDB Atlas — measured 40% faster page loads via SSR and achieved 95+ Lighthouse scores.
- Implemented secure JWT authentication, bcryptjs password hashing, and Markdown CMS; served as a reusable template for 3+ other developers.

AI-Powered Crop Yield Prediction

 GitHub

Python, XGBoost, SHAP, Streamlit

- Engineered end-to-end ML pipeline using XGBoost, achieving 92% prediction accuracy on a held-out test set for Rice/Wheat/Maize yields through feature engineering and hyperparameter tuning.
- Integrated SHAP for model explainability and deployed interactive Streamlit dashboard, processing 10,000+ data points and reducing manual analysis time by 70%.

DrishtiLens: ISL Neural Translation (Ongoing Research)

Python, OpenCV, TensorFlow

- Spearheaded real-time Indian Sign Language translation — built frame-by-frame pipeline with dataset curation across 100,000+ gesture samples, achieving 85% baseline accuracy on validation set.
- Investigating MobileNet architecture and latency optimization for edge-compute deployment, targeting sub-100ms inference on resource-constrained hardware.